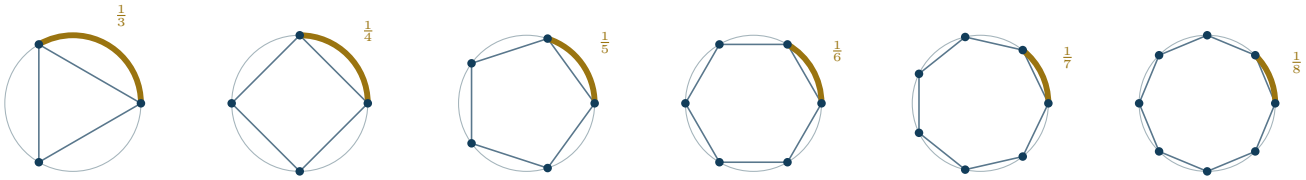
CYCLOTOMIC FIELDS ON ONE PAGE

Cyclotomic fields and the arcs of inscribed polygons

The vertex of a regular n-gon is $\zeta_n = e^{2\pi i/n}$, one arc of $\frac{1}{n}$ of the circle, and it generates the field $\mathbb{Q}(\zeta_n)$.

1 / THE RULE AND THE PICTURE

Each step round the inscribed n-gon is an arc of $\frac{1}{n}$ of the circle, multiplication by ζ_n



$n=3$ $n=4$ $n=5$ $n=6$ $n=7$ $n=8$

The roots of $x^n = 1$ are the vertices ζ_n^k, the smallest field containing ζ_n is the cyclotomic field $\mathbb{Q}(\zeta_n)$, and its degree over $\mathbb{Q}$ is Euler's totient $\varphi(n)$:

$$\zeta_n = e^{2\pi i/n}, \qquad [\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n), \qquad \text{ring of integers } \mathbb{Z}[\zeta_n]$$

2 / THE GEOMETRY / ARC AND CHORD

The arc shrinks as $2\pi/n$ and the chord $|1-\zeta_n| = 2\sin\frac{\pi}{n}$ follows it from below

n	arc fraction	angle	arc length $2\pi/n$	chord $2\sin\frac{\pi}{n}$	chord/arc	product of chords
3	1/3	120°	2.094	$\sqrt{3} = 1.732$	0.827	$\sqrt{3}\cdot\sqrt{3} = 3$
4	1/4	90°	1.571	$\sqrt{2} = 1.414$	0.900	$\sqrt{2}\cdot 2\cdot\sqrt{2} = 4$
5	1/5	72°	1.257	1.176	0.935	$1.176^2\cdot 1.902^2 = 5$
6	1/6	60°	1.047	1	0.955	$1\cdot\sqrt{3}\cdot 2\cdot\sqrt{3}\cdot 1 = 6$
7	1/7	51.43°	0.898	0.868	0.967	7
8	1/8	45°	0.785	0.765	0.974	8

The product of the $n-1$ chords from one vertex to the others is always n, because $\prod_{k=1}^{n-1}(1-\zeta_n^k)$ is $\frac{x^n-1}{x-1}$ evaluated at $x=1$.

3 / THE ALGEBRA / WHICH FIELD EACH POLYGON GENERATES

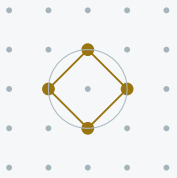
Degree $\varphi(n)$ is 2 for $n = 3, 4, 6$ and larger for $5, 7, 8$

n	$\Phi_n(x)$	$\varphi(n)$	field and ring	a familiar subfield	$\Phi_n(1)$	compass
3	x^2+x+1	2	$\mathbb{Q}(\omega)$, $\mathbb{Z}[\omega]$ Eisenstein	itself	3	yes
4	x^2+1	2	$\mathbb{Q}(i)$, $\mathbb{Z}[i]$ Gaussian	itself	2	yes
5	$x^4+x^3+x^2+x+1$	4	$\mathbb{Q}(\zeta_5)$	$\mathbb{Q}(\sqrt{5})$, golden ratio	5	yes
6	x^2-x+1	2	$\mathbb{Q}(\omega)$, the same as $n=3$	itself	1	yes
7	$x^6+x^5+\cdots+1$	6	$\mathbb{Q}(\zeta_7)$	$\mathbb{Q}(\sqrt{-7})$ and a real cubic	7	no
8	x^4+1	4	$\mathbb{Q}(\zeta_8) = \mathbb{Q}(i,\sqrt{2})$	$\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$	2	yes

The value $\Phi_n(1)$ is the norm of $1-\zeta_n$, so $1-\zeta_6$ is a unit (its chord is the radius) while $1-\zeta_3$ and $1-\zeta_4$ have norms 3 and 2. Gauss showed that the polygon can be built with ruler and compass exactly when $\varphi(n)$ is a power of 2, which excludes only the heptagon here.

4 / GAUSSIAN AND EISENSTEIN INTEGERS / THE TWO LATTICES

The square arc $\frac{1}{4}$ gives $\mathbb{Z}[i]$ and the sixth arc $\frac{1}{6}$ gives $\mathbb{Z}[\omega]$, and their units are the polygons



$\mathbb{Z}[i]$: units $\pm 1, \pm i$



$\mathbb{Z}[\omega]$: six units

In $\mathbb{Z}[i]$ the norm is a^2+b^2 and in $\mathbb{Z}[\omega]$ it is a^2-ab+b^2. A prime p splits according to $p \bmod n$: in $\mathbb{Z}[i]$ the prime $5 \equiv 1 \pmod 4$ splits as $(2+i)(2-i)$, 3 stays prime and $2 = -i(1+i)^2$ ramifies; in $\mathbb{Z}[\omega]$ the prime $7 \equiv 1 \pmod 3$ splits as $(3+\omega)(3+\omega^2)$ with norm $9-3+1=7$, while 2 and 5 stay prime and 3 ramifies.

5 / BEYOND THE LATTICES / UNITS AND THE CRYSTALLOGRAPHIC RESTRICTION

When $\varphi(n) > 2$ the ring is dense in the plane, and the extra units are the metallic means

For $n = 5, 7, 8$ the ring $\mathbb{Z}[\zeta_n]$ has rank $4, 6, 4$ over $\mathbb{Z}$ and fills the plane densely instead of forming a lattice, so the arcs $\frac{1}{5}, \frac{1}{7}, \frac{1}{8}$ cannot be rotation symmetries of a crystal, while $\frac{1}{3}, \frac{1}{4}, \frac{1}{6}$ can ($\varphi(n) \le 2$); the five-fold and eight-fold cases appear in quasicrystals. Their unit groups are infinite: $n=5$ has the golden ratio $\varphi = -(\zeta_5^2+\zeta_5^3) = 1.618$, $n=8$ has the silver ratio $1+\zeta_8+\zeta_8^{-1} = 1+\sqrt{2} = 2.414$, and $n=7$ has $2\cos\frac{2\pi}{7} = 1.247$, a root of t^3+t^2-2t-1, while for $n = 3, 4, 6$ the only units are the roots of unity themselves.

ASIDE / THE SAME FIELD TWICE

Since $-\omega^2 = \zeta_6$ the fields for $n=3$ and $n=6$ coincide, and in general $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{2n})$ for odd n.

ASIDE / CHECK IN PYTHON

`sympy.cyclotomic_poly(8,x)` gives x^4+1, and `prod(2*sin(pi*k/7))` over $k = 1..6$ gives 7.

Convention. $\omega = e^{2\pi i/3}$, $\varphi(n)$ is Euler's totient, Φ_n the cyclotomic polynomial. After the cyclotomic fields project and the earlier sheets on the totient and the Legendre symbol.

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